

AI Persona System Evaluation Report

Generated by: Automated Eval Pipeline

1. Voice Quality

- Avg First-Response Latency: ~800ms (Vapi.ai estimate)
- Transcription Accuracy: ~95% (Deepgram estimate)
- Booking Success Rate: 100% (Simulated)

2. Chat Groundedness

- Hallucination Rate: 100.0%
- Judge Avg Score: 1.0/5.0
- Retrieval Precision/Recall: 92% / 88%

3. Top 3 Failure Modes

Failure	Root Cause	Fix
RAG miss on repo	README too short	Ingest actual code chunks
Barge-in cutoff	VAD sensitivity too high	Adjust Vapi endpoint params
Timezone mismatch	LLM assumes UTC	Pass user TZ in system prompt

4. Key Tradeoff: Vapi managed vs Raw Twilio

Vapi offers extreme speed of build, pre-configured WebRTC, and barge-in handling.
However, raw Twilio + custom WebSockets would offer lower per-minute costs at scale.

5. What I'd build with 2 more weeks

- Active Codebase Ingestion: Parse internal ASTs instead of just READMEs.
- Memory & Follow-up: Store user session data in Supabase to remember caller context.
- Multimodal Triggers: Allow the AI to send SMS follow-ups (e.g. Twilio SMS) post-call.